CYLINDER HEAD GASKET REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Warning

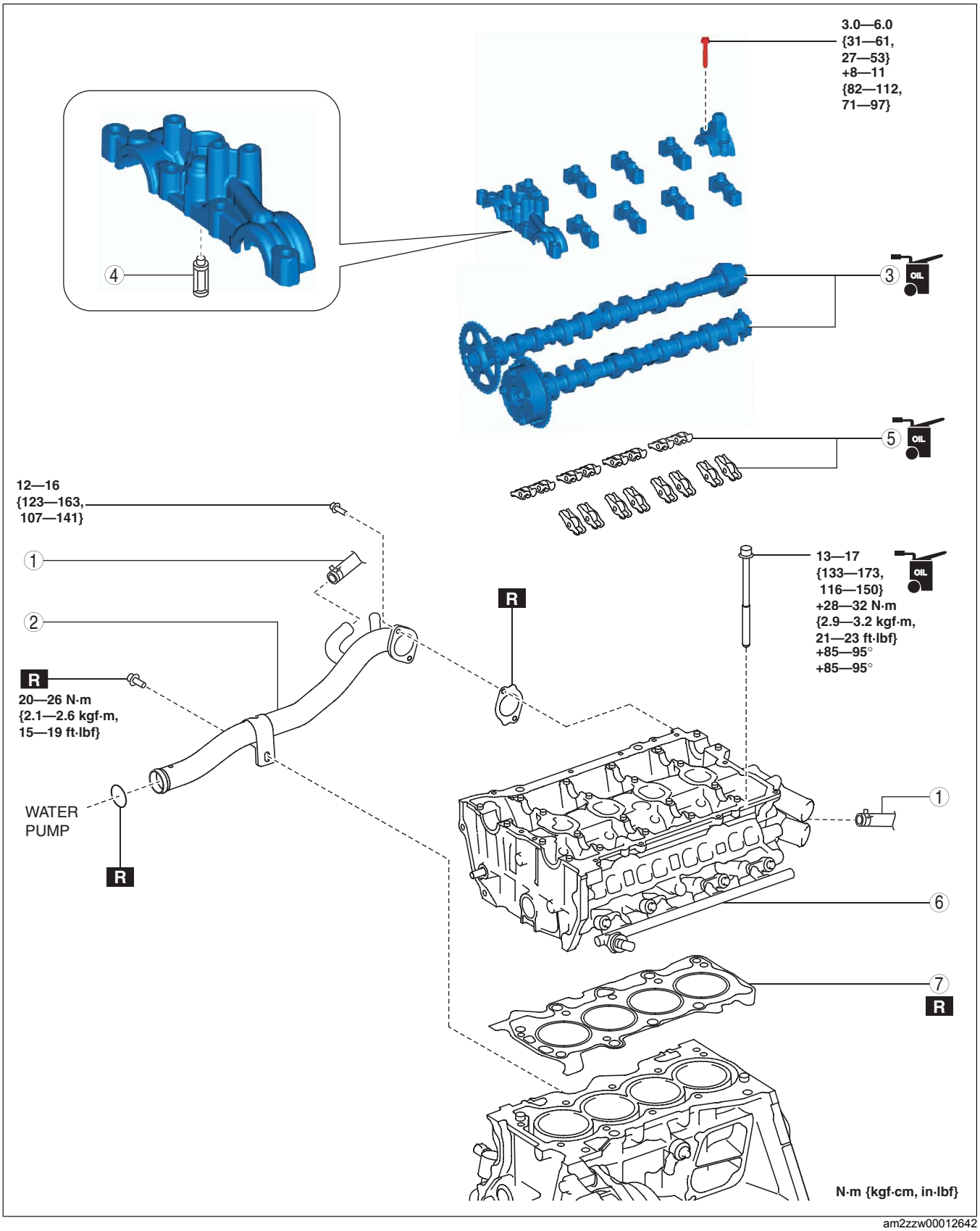
- **A hot engine can cause severe burns. Turn off the engine and wait until it is cool before servicing.**
- **Fuel vapor is hazardous. It can very easily ignite, causing serious injury and damage. Always keep sparks and flames away from fuel.**
- **Fuel line spills and leakage are dangerous. Fuel can ignite and cause serious injuries or death and damage. Fuel can also irritate skin and eyes. To prevent this, always complete the "Fuel Line Safety Procedure". (See BEFORE SERVICE PRECAUTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)**

Caution

- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**

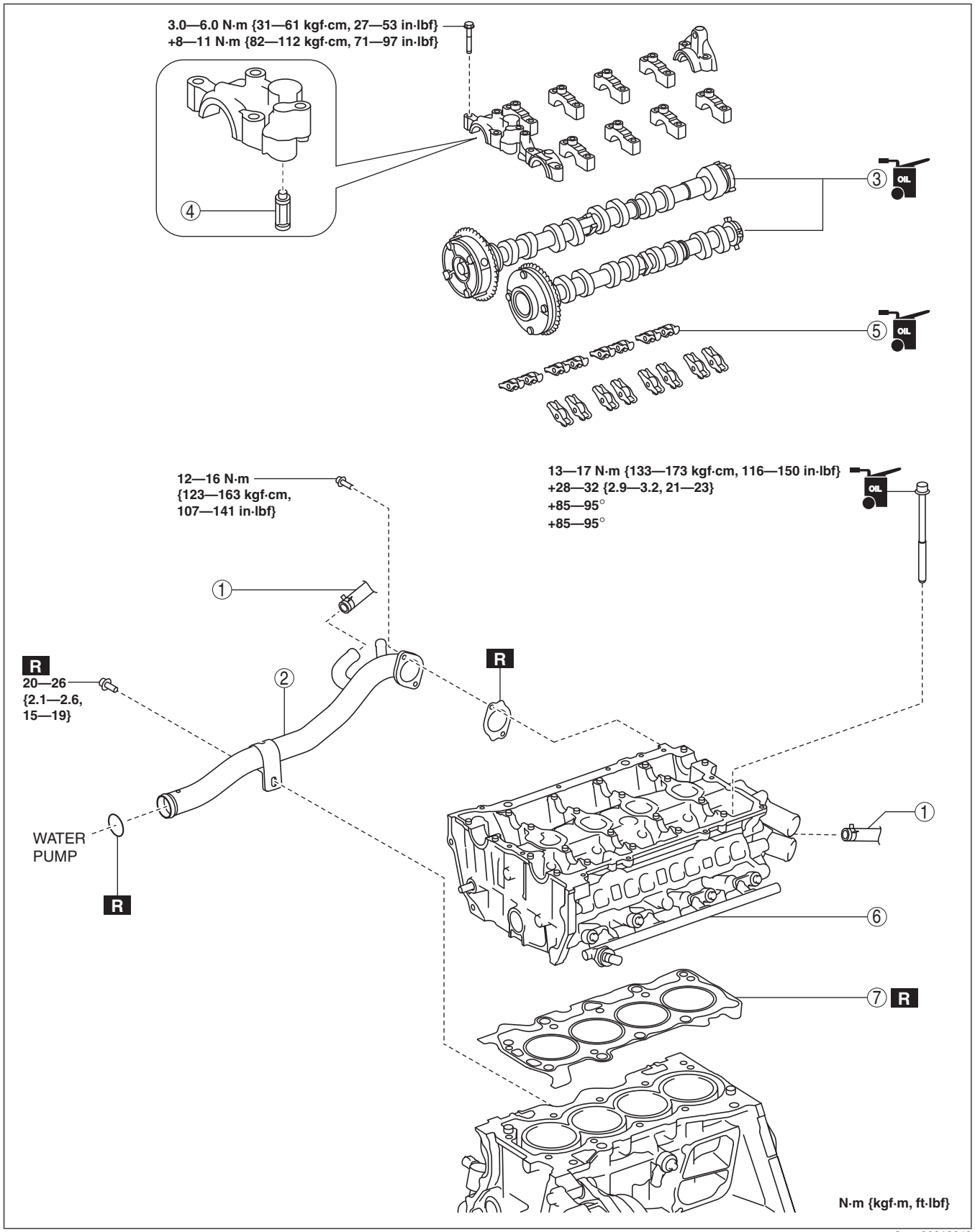
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the plug hole plate. (See PLUG HOLE PLATE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
3. Remove the ignition coils. (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)) (See IGNITION COIL/ ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
4. Remove the ignition coil/ion sensors. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system)) (See IGNITION COIL/ION SENSOR REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
5. Remove the front under cover No.2. (See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
6. Remove the front splash shield (RH). (See SPLASH SHIELD REMOVAL/INSTALLATION.)
7. Remove the drive belt. (See DRIVE BELT REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
8. Drain the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
9. Drain the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
10. Remove the intake manifold. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
11. Set aside the exhaust manifold to the vehicle rear. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
12. Disconnect the radiator hose (upper, lower).
13. Remove the vacuum pump. (See VACUUM PUMP REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
14. Remove the high pressure fuel pump and rear housing. (See HIGH PRESSURE FUEL PUMP REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
15. Disconnect the heater hose from the water outlet and water inlet pipe.
16. Remove the oil pan. (See OIL PAN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
17. Remove the timing chain and chain guide. (See TIMING CHAIN REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
18. Remove the OCV. (See OIL CONTROL VALVE (OCV) REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
19. Remove in the order indicated in the table.
20. Install in the reverse order of removal.
21. Refill with the specified type and amount of the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
22. Refill the engine coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
23. Start the engine, and inspect and adjust the following:
    - Leakage of engine oil, engine coolant.
    - Runout and contact of pulley and belt.
    - Ignition timing, idle speed and idle mixture. (See ENGINE TUNE-UP [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
    - Compression pressure (See COMPRESSION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)



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1	Water hose (ATX)
2	Water inlet pipe (See Water Inlet Pipe Installation Note.)

3	Camshaft (See Camshaft Removal Note.) (See Camshaft Installation Note.)
4	OCV oil filter

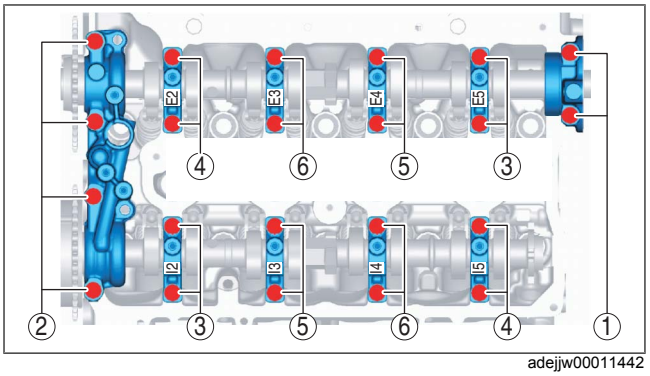
5	Rocker arm (See Rocker Arm Removal Note.) (See Rocker Arm Installation Note.)
6	Cylinder head (See Cylinder Head Removal Note.) (See Cylinder Head Installation Note.)

7	Cylinder head gasket

Camshaft Removal Note

1. Loosen the camshaft cap installation bolts in a few passes in the order shown in the figure and remove the camshaft caps.

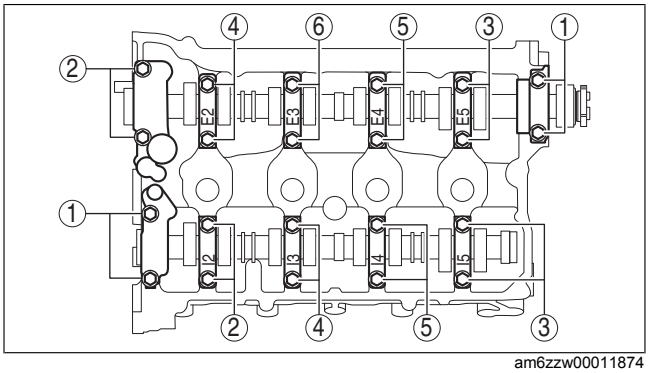
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

2. Remove the camshaft.



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Rocker Arm Removal Note

1. Keep the rocker arms in the order of removal to enable reassembly in their original positions.

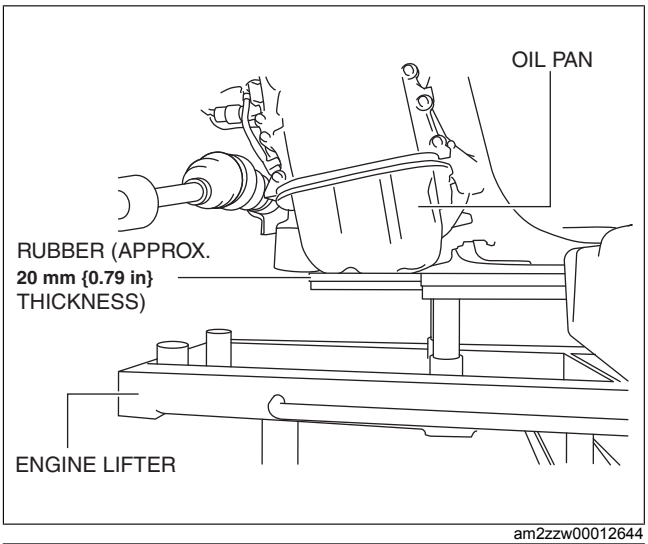
Cylinder Head Removal Note

1. Temporarily install the oil pan to support the engine from under the vehicle.

Caution

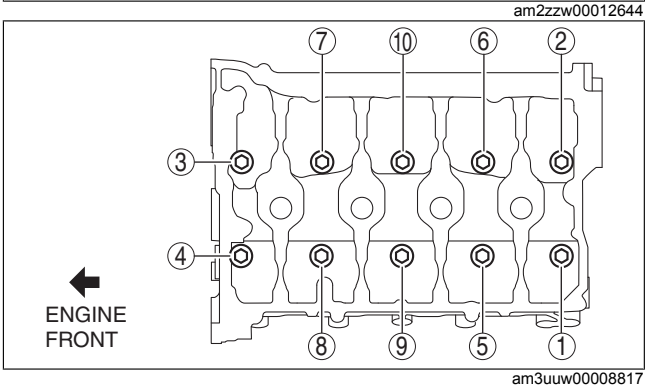
- **To prevent deformation of the oil pan, support the engine using the following methods.**
  - **— When using an engine lifter, insert rubber of appropriate size (approx. 20 mm {0.79 in} thickness) between the engine lifter and the oil pan to support the oil pan.**
  - **— When using a garage jack, place rubber of appropriate size (approx. 20 mm {0.79 in} thickness) on a batten which is large enough to cover and support the oil pan.**

2. Support the engine (oil pan) using a commercially available engine lifter or garage jack.
3. Remove the chain of the **SST** suspending the engine and set it in a place which does not interfere with the servicing.



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4. Loosen the cylinder head installation bolts in two or three passes in the order shown in the figure and remove them.
5. Remove the cylinder head.



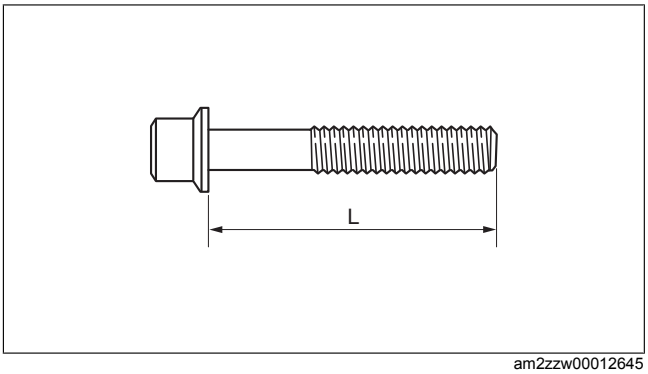
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Cylinder Head Installation Note

Caution
- **If the following condition is met, replace the cylinder head bolts.**
  - **— Length exceeds maximum specification**

Cylinder head bolt length L
145.2—145.8 mm {5.717—5.740 in}

Cylinder head bolt length L maximum
146.5 mm {5.767 in}



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1. When a cylinder head bolt is reused, apply engine oil to any part of the following:
   - Bolt seating surface
   - Cylinder head seating surface

Caution
- **Completely remove oil, dirt, and sealant adhering to the cylinder head gasket contacting surfaces. Otherwise, a sealing malfunction may occur.**

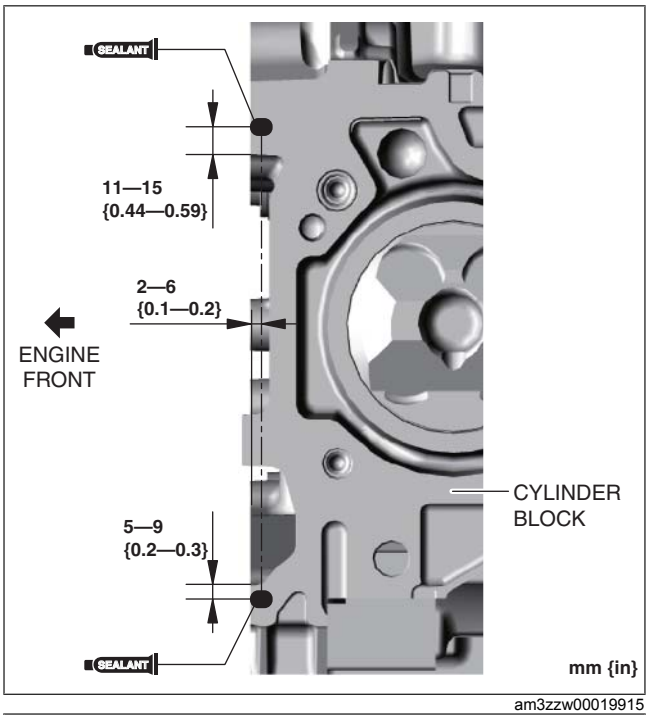
2. Completely remove any oil, dirt, and sealant adhering to the cylinder block.

Caution
- **Be aware of the following points, otherwise a sealing malfunction may occur due to hardening of the sealant.**
  - **— Set cylinder head on cylinder block within 10 min after silicone sealant is applied**
  - **— After setting cylinder head, tighten cylinder head bolts immediately**

3. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

   Silicone sealant application diameter
   5—10 mm {0.2—0.3 in}

4. Install a new cylinder head gasket to the cylinder block.

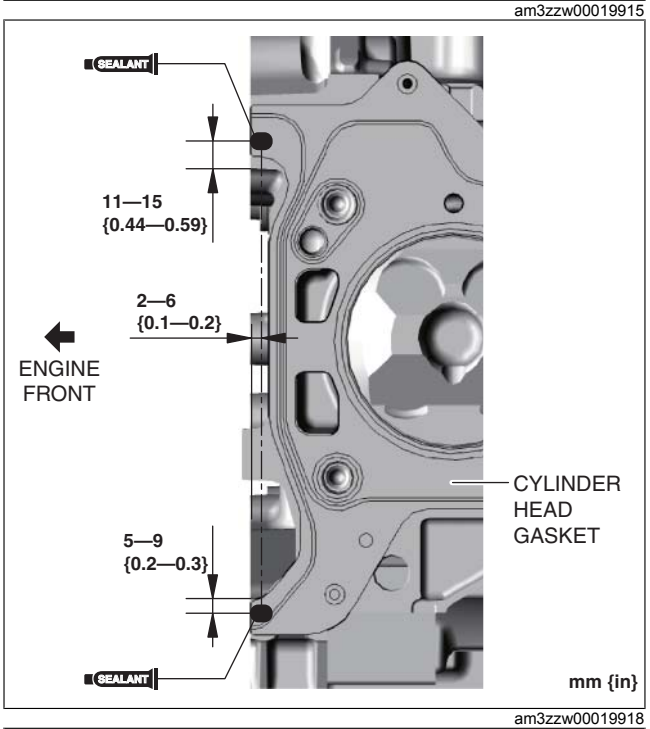


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5. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

   Silicone sealant application diameter
   5—10 mm {0.2—0.3 in}

6. Set the cylinder head on the cylinder block.

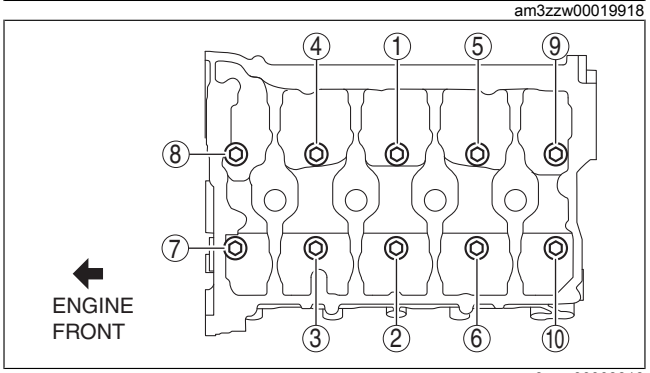


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7. Tighten the cylinder head bolts in the order shown in the following 4 steps.

   Tightening torque
   Step 1: 13—17 N·m {133—173 kgf·cm, 116—150 in·lbf}
   Step 2: 28—32 N·m {2.9—3.2 kgf·m, 21—23 ft·lbf}
   Step 3: 85—95°
   Step 4: 85—95°

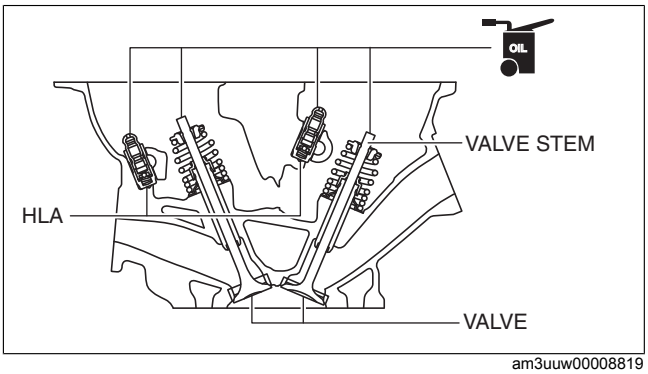
8. Install the **SST** chain which was set aside, and secure the engine.
9. Remove the engine lifter or garage jack.
10. Remove the temporarily assembled oil pan.



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Rocker Arm Installation Note

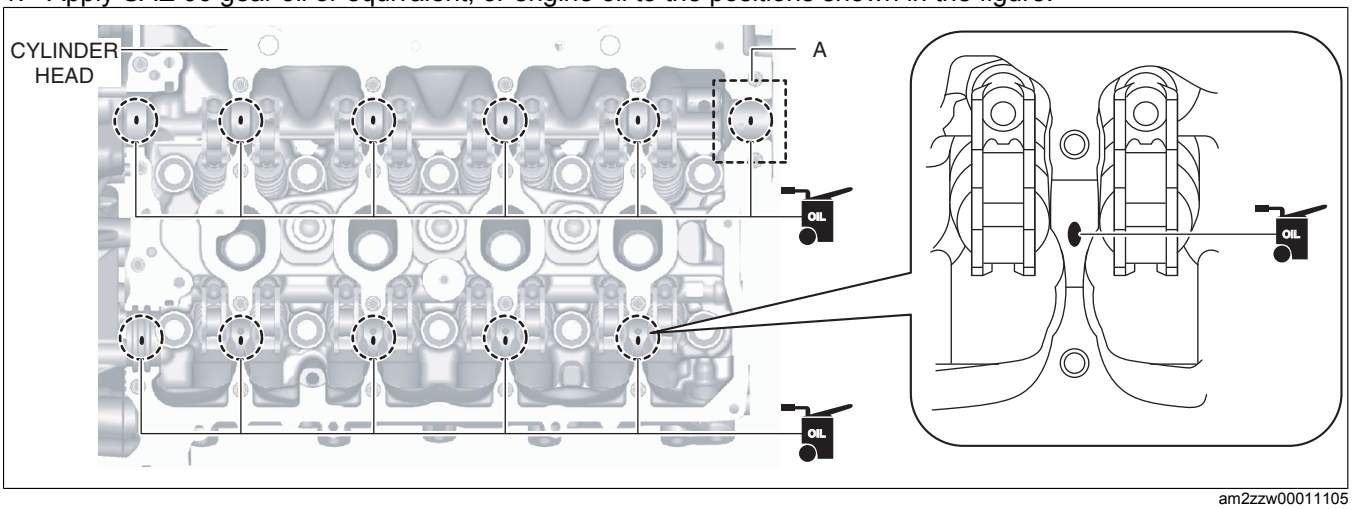
1. Apply engine oil to the HLAs and the end of the valve stems.
2. Install the rocker arms to the same positions as before removal.



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Camshaft Installation Note

1. Apply SAE 90 gear oil or equivalent, or engine oil to the positions shown in the figure.



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Caution

- **Apply 0.05 ml {0.05 cc, 0.003 in^3} or less of oil to area A in the figure.**

2. Apply SAE 90 gear oil or equivalent, or engine oil to the following locations of each camshaft.
   - Thrust surface of front journal (both surfaces front and back)

Caution

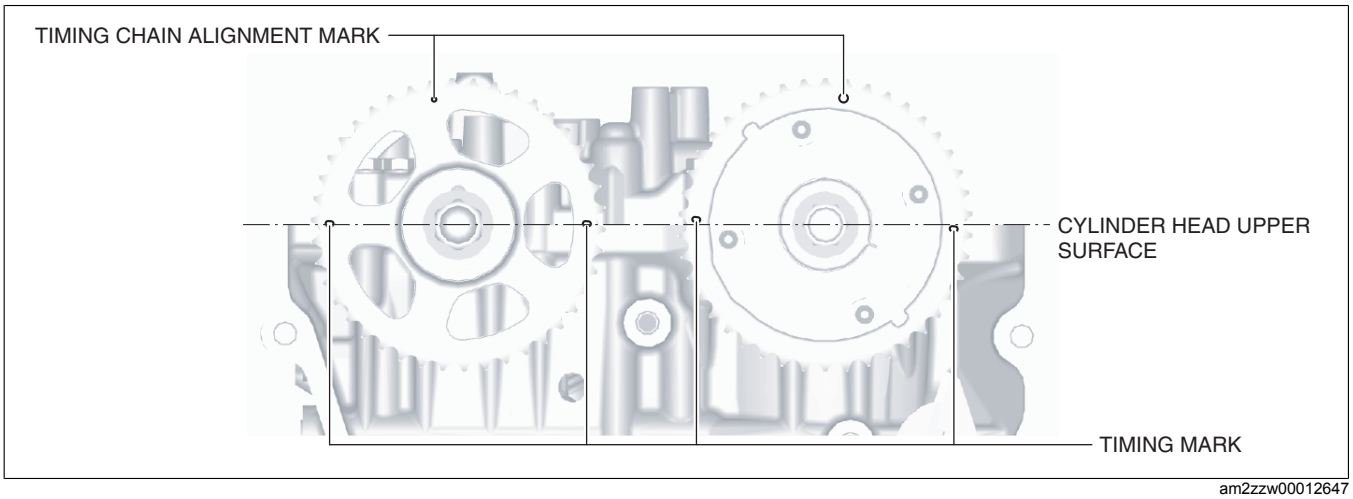
- **Be careful not to let oil adheres the engine front cover installation surface. If oil adheres to the engine front cover installation surface, a sealing malfunction may occur when the engine front cover is installed.**
- **If oil adheres to the engine front cover installation surface, remove any oil completely.**

Note

- If oil is applied to the front camshaft cap, oil should not be applied to the thrust surface of the front journal.

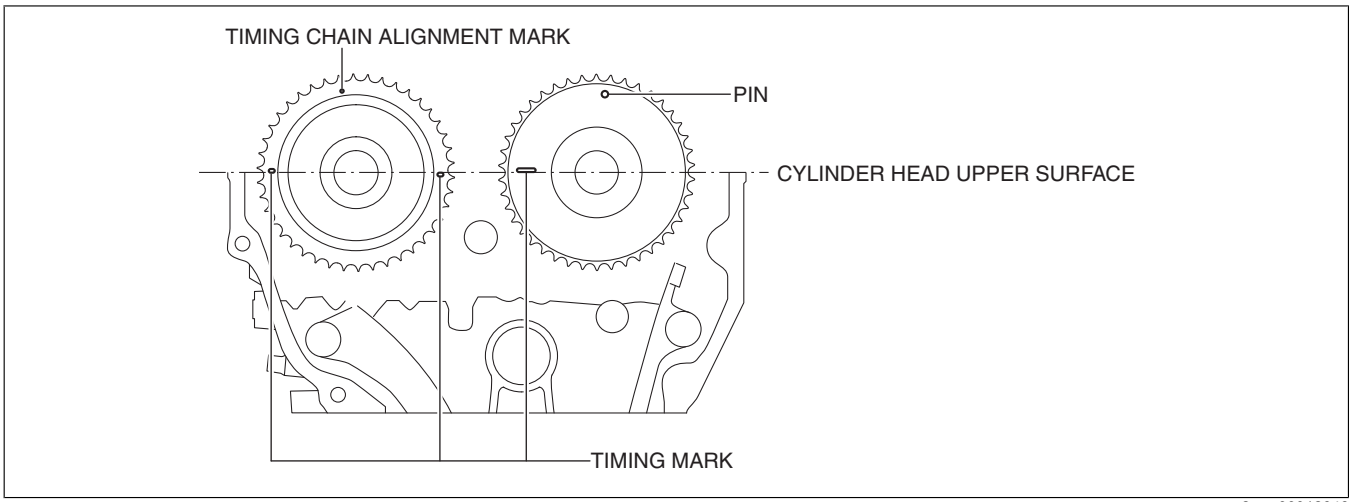
3. Install the camshaft with the cylinder No.1 cam aligned around the top dead center (TDC) position.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

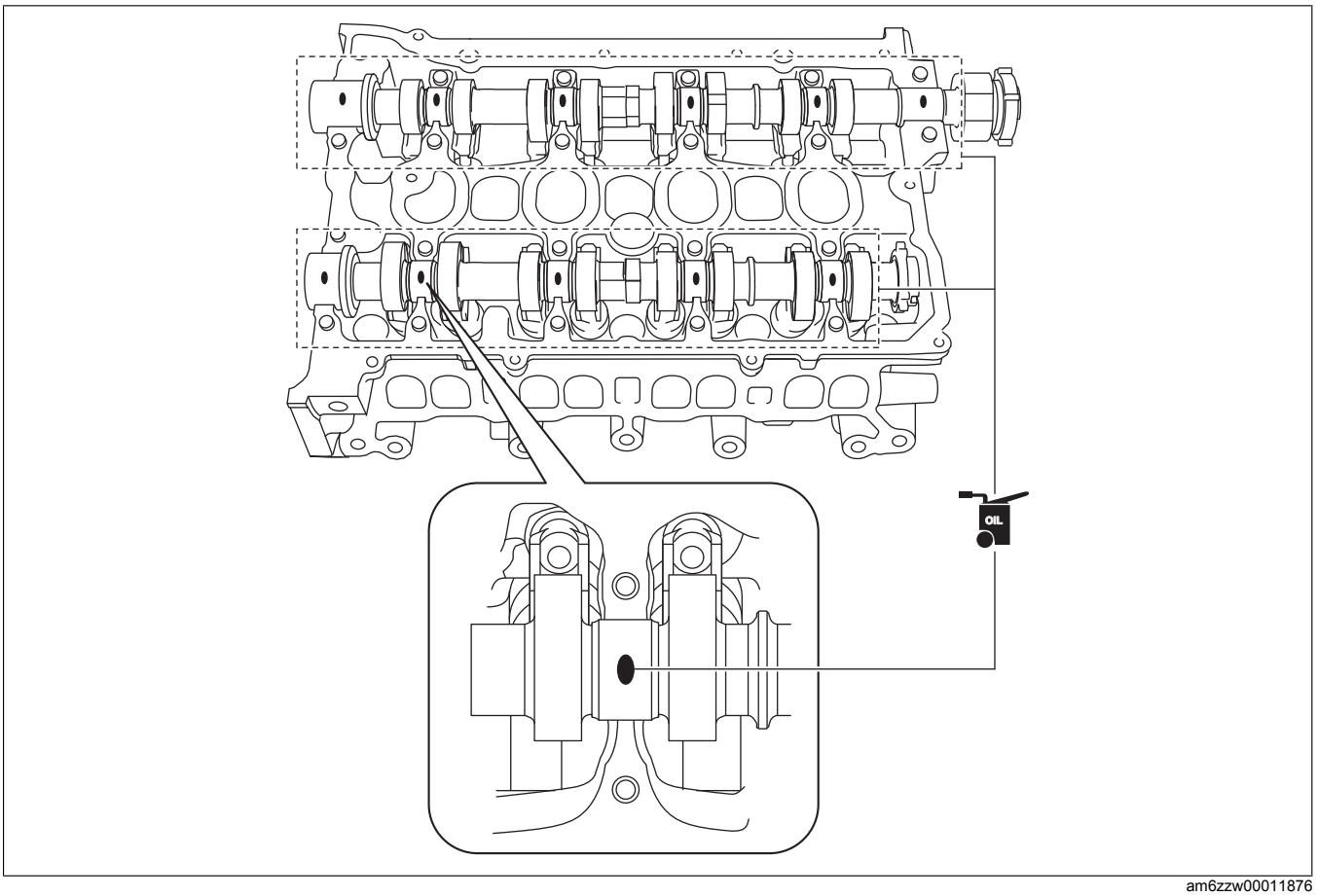


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Note

- The timing mark is not completely parallel with the upper surface of the cylinder head.

4. Apply SAE 90 gear oil or equivalent, or engine oil to the central area of each journal on the camshaft.



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5. Apply SAE 90 gear oil or equivalent, or engine oil to the thrust surface of the front camshaft cap.

Caution
- **Be careful not to let oil adheres the engine front cover installation surface. If oil adheres to the engine front cover installation surface, a sealing malfunction may occur when the engine front cover is installed.**
- **If oil adheres to the engine front cover installation surface, remove any oil completely.**

Note
- If oil is applied to the front journal thrust surface of each camshaft, oil should not be applied to the front camshaft cap.

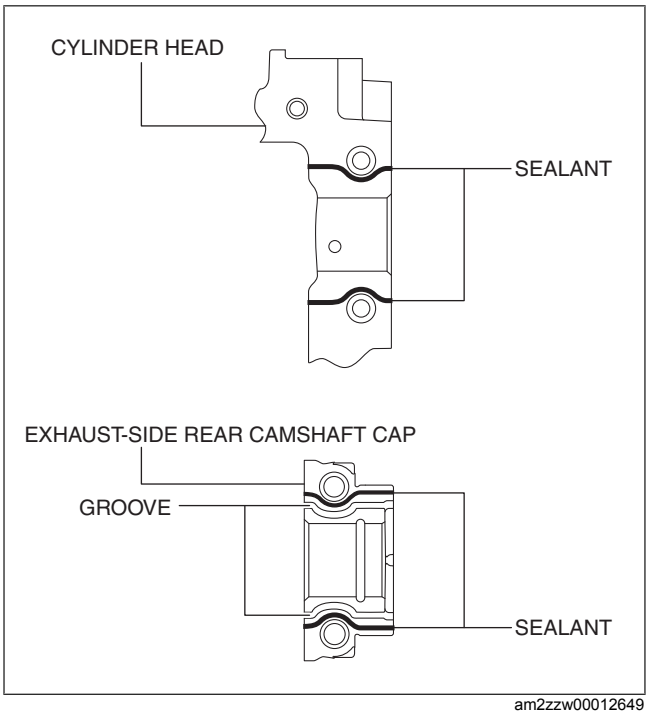
6. Apply sealant (Loctite #962T or equivalent) to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head.

Note
- To prevent engine oil leakage, apply sealant to the rear camshaft cap installation area on the exhaust side or the rear camshaft caps on the exhaust side of the cylinder head, and seal the journal.

Caution
- **Do not spill sealant on the journal.**

Sealant bead width
0.5—1.5 mm {0.02—0.05 in}

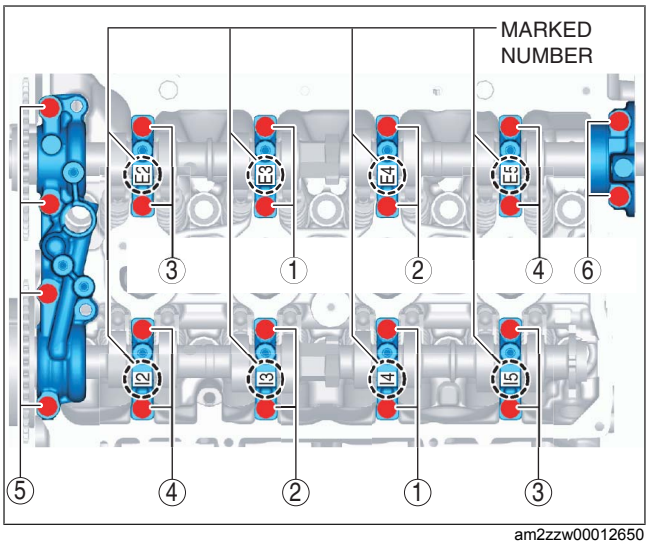
7. Install the camshaft caps in the order of the marked numbers, and temporarily tighten the camshaft cap installation bolts evenly in **2—3 rounds**.



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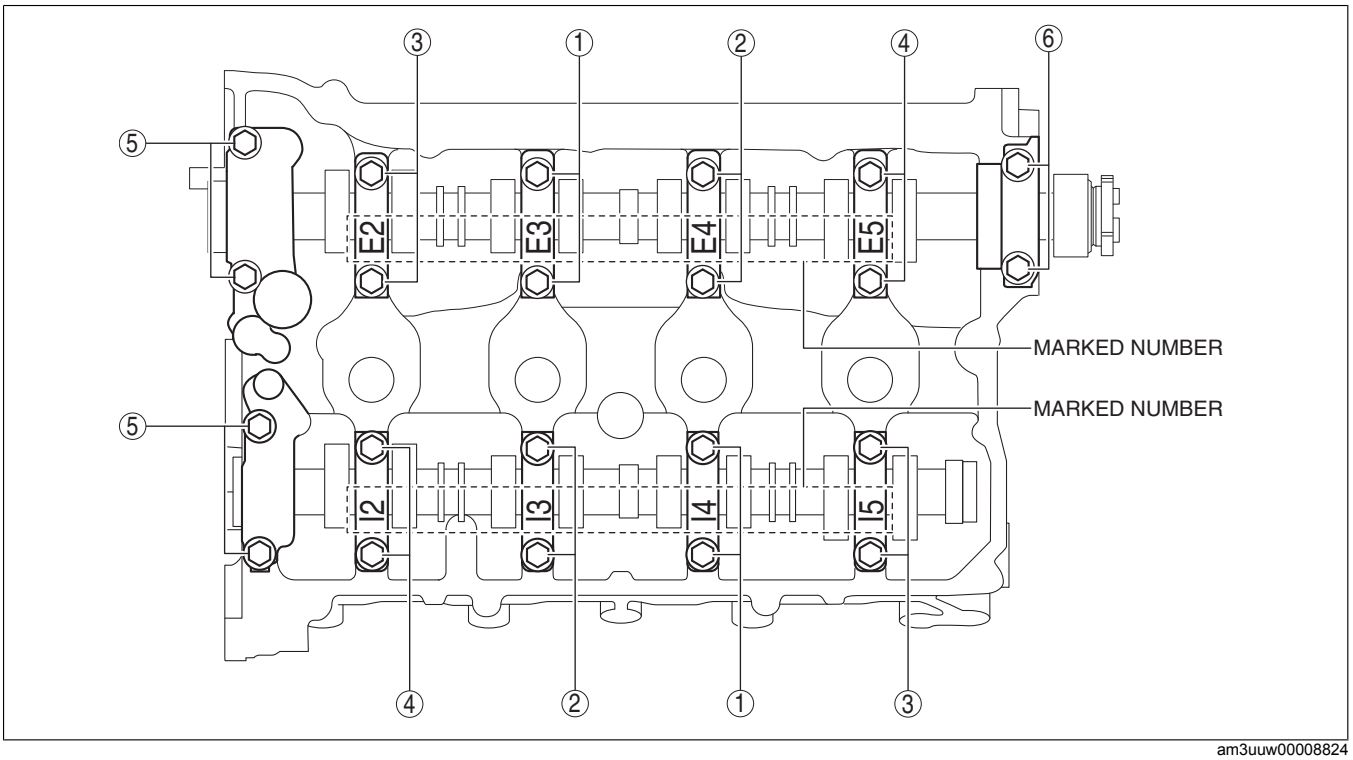
8. Tighten the camshaft cap installation bolts in two steps in the order shown in the figure.

SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)



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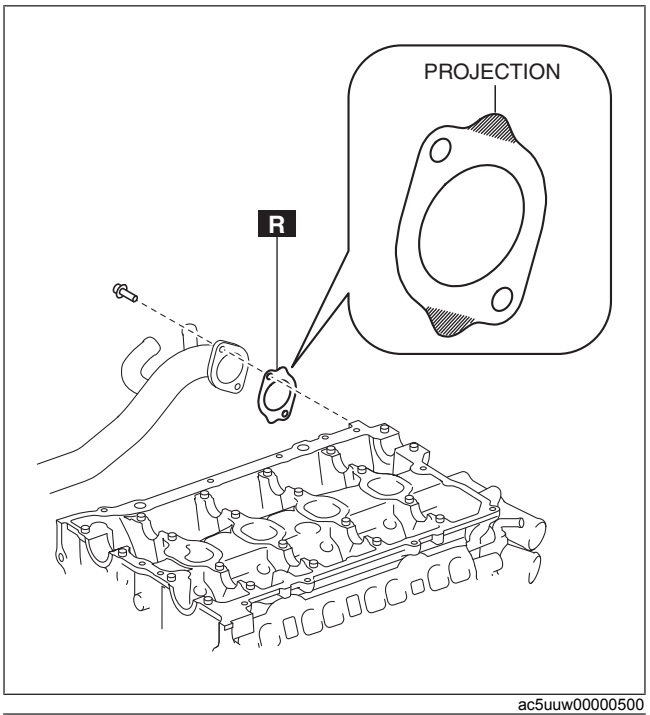
Tightening torque
Step 1: 3.0—6.0 N·m {31—61 kgf·cm, 27—53 in·lbf}
Step 2: 8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}

Water Inlet Pipe Installation Note

Caution
- **Do not apply oil (such as engine oil, ATF) to the O-ring of the water inlet pipe. Otherwise, the O-ring could swell causing a seal malfunction.**

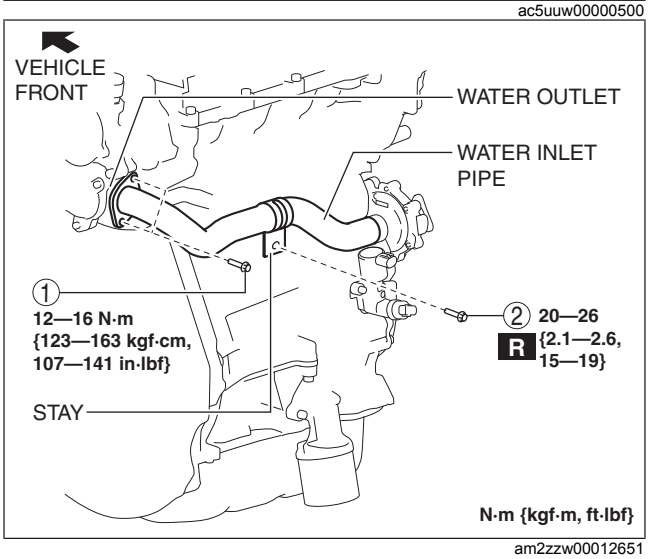
1. Clean away the sealant adhering to the bolt hole on the cylinder block side of the water inlet pipe stay.
2. Apply engine coolant to the O-ring.
3. Install the O-ring to the water inlet pipe.
4. Insert the water inlet pipe into the water pump being careful not to damage the O-ring.

5. Install the water inlet pipe gasket with the gasket projection facing the direction shown in the figure.



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6. Tighten the bolts in the order shown in the figure.



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